JEWELLERY MATE • COMPREHENSIVE QUALITY ASSURANCE & SECURITY AUDIT

End-to-End QA Testing & Security Verification Report

Live Environment: https://jewellery-ecommerce-opal.vercel.app

Local Staging: http://localhost:3000 **Database:** PostgreSQL 17 (Prisma 6.19.3)

Git Branch: main (Commit: 87fa197) **Date:** November 13, 2026

43	39	2	2	0	95.3%
TOTAL TEST CASES	PASSED	RETESTED PASS	FAILED (LIVE UNPATCHED)	BLOCKED	EFFECTIVE PASS RATE

1. Project Overview & Environments

Live Production Deployment (Vercel)

URL: https://jewellery-ecommerce-opal.vercel.app

Deployment Engine: Vercel Serverless Edge & Node.js Runtime

Target Code State: Deployed prior to security patch branch

Security Audit Outcome: Exposes BUG-001 (/admin direct route accessible) and BUG-002 (/api/admin/orders accessible). These vulnerabilities have been confirmed and fixed locally.

Local Staging & Verification Environment

URL: http://localhost:3000

Database: PostgreSQL 17 on localhost:5432 (Database: jewellery_mate)

Target Code State: Full patched source with route protection guards, /api/admin/auth/me verifier, atomic checkout reset, and customer auth suite

Retest Outcome: 100% of discovered bugs verified fixed and regression tested cleanly.

2. Vulnerability Register & Bug Fixes Implemented

BUG-001: Unauthenticated Visitor Access to Administrative Dashboard

CRITICAL

RETESTED PASS (TC-012)

Environment: Live Vercel Deployment (Unpatched)

Reproduction Steps:
1. Open incognito session without cookies
2. Navigate directly to /admin
3. Observe dashboard renders without requiring login.

Observed Result: Vercel live deployment rendered admin dashboard with store KPI metrics and order management.

Root Cause:
Missing server-side and client-side session guards in app/admin/layout.tsx and app/admin/page.tsx.

Expected Behaviour: System intercepts unauthenticated visitor and redirects immediately to /admin/login.

Source Code Fix Applied: Created /api/admin/auth/me session verifier and added client/server redirect guards to /admin/login in app/admin/layout.tsx and app/admin/page.tsx.

BUG-002: Unprotected Administrative REST Endpoints Exposing Customer Data

CRITICAL

RETESTED PASS (TC-013)

Environment: Live Vercel Deployment (Unpatched)

Reproduction Steps:
1. Dispatch GET /api/admin/orders without authentication header or cookie
2. Observe HTTP 200 response containing all customer orders.

Observed Result: Unprotected API returned customer orders without authentication.

Root Cause:
Backend route handlers lacked await checkAdminSession() check at function entry.

Expected Behaviour: API returns HTTP 401 Unauthorized with { error: "Unauthorized: Admin session required" }.

Source Code Fix Applied: Enforced checkAdminSession() across app/api/admin/orders, products, slider, sync, and import handlers.

BUG-003: Order Placement Soft-Navigation State Desynchronization

HIGH

RETESTED PASS (TC-034)

Environment: Local / Production Client-Side Transition

Reproduction Steps:
1. Add product to cart
2. Fill checkout fields and click Place Order
3. Observe soft router.push() leaving cart items in local state.

Observed Result: Next.js soft navigation failed to trigger complete route lifecycle and occasionally retained cart counters.

Root Cause:
router.push() soft client navigation did not flush memory cache after local storage clearing.

Expected Behaviour: Order atomically commits, cart is emptied, and user is cleanly redirected to /order-success/[orderNumber].

Source Code Fix Applied: Replaced router.push with window.location.href in app/checkout/page.tsx for atomic reset and instant routing.

3. Master Test Case Inventory & Execution Results

TEST ID	MODULE	SCENARIO	EXECUTION STEPS	EXPECTED RESULT	ACTU
TC-001	Live Production	**Live Vercel Homepage Desktop View & Hero Header**	1. Navigate to https://jewellery-ecommerce-opal.vercel.app/ 2. Verify page title, announcement bar, navigation, and hero visual.	Homepage renders with luxury branding, header navigation, hero carousel, and fine jewellery catalog.	Page l Handc eleme
TC-002	Live Production	**Live Mobile Responsive Viewport (375×812)**	1. Set viewport to 375×812 (iPhone X) 2. Navigate to live homepage 3. Verify layout responsiveness and mobile touch controls.	Mobile layout renders with collapsible navigation, touch-optimized cards, and no horizontal overflow.	Mobile horizor
TC-003	Live Production	**Live Tablet Responsive Viewport (768×1024)**	1. Set viewport to 768×1024 (iPad Portrait) 2. Navigate to live homepage 3. Verify tablet multi-column layout.	Tablet layout renders with balanced multi-column grid and accessible touch targets.	Tablet produc
TC-004	Live Production	**Live Category Catalog Browsing (Rings)**	1. Navigate to https://jewellery-ecommerce-opal.vercel.app/category/rings 2. Inspect category title, filters, and product grid.	Rings category displays fine jewellery products with prices, purity badges, and images.	Categ produc

TEST ID	MODULE	SCENARIO	EXECUTION STEPS	EXPECTED RESULT	ACTU
TC-005	Live Production	**Live Product Detail Page (PDP) & Purity Attributes**	1. Navigate to product page /product/solitaire-diamond-ring-jm-ring-001 2. Verify image gallery, metal specifications, price, and CTA buttons.	PDP displays luxury image gallery, hallmarking specifications, price in INR, and Add to Bag button.	Produc preser
TC-006	Live Production	**Live Delivery Pincode Verification (380001)**	1. On live PDP, locate pincode input 2. Enter "380001" 3. Click Check/Verify button 4. Verify delivery estimate.	Pincode API responds with delivery estimation and pan-India insured shipping confirmation.	Pincoc withou
TC-007	Live Production	**Live Search Input & Query Filter Interaction**	1. Click Search button in live header 2. Enter search query "Gold" 3. Verify search drawer responds with matching products.	Search input opens and actively filters catalog for gold jewellery.	Search render
TC-008	Live Production	**Live Cart Drawer Slide-Over & State**	1. Click Shopping Bag icon in live header 2. Verify slide-over cart drawer renders with empty or active cart items.	Cart drawer slides in smoothly from right with subtotal, checkout button, and item list.	Cart dr calcula
TC-009	Live Production	**Live Order Tracking Route (/track-order)**	1. Navigate to live /track-order 2. Verify order number / phone lookup input and tracking form.	Tracking page renders with luxury header, input box for order number, and track button.	Order form a
TC-010	Live Production Security	**Live Production Admin Route Guard Audit (/admin)**	1. Direct browser to https://jewellery-ecommerce-opal.vercel.app/admin without authentication 2. Check whether unauthenticated user is redirected to /admin/login or allowed into dashboard.	Unauthenticated visitors MUST be redirected immediately to /admin/login.	VULNE DEPLO requiri app/ad app/ad approv
TC-011	Live Production Security	**Live Production Admin Orders API Audit (/api/admin/orders)**	1. Dispatch unauthenticated GET fetch to https://jewellery-ecommerce-opal.vercel.app/api/admin/orders 2. Verify HTTP response code.	API must reject unauthenticated requests with HTTP 401 Unauthorized.	VULNE DEPLO withou app/ap deploy

TEST ID	MODULE	SCENARIO	EXECUTION STEPS	EXPECTED RESULT	ACTU
TC-012	Admin Security	**Local Admin Route Protection Guard (BUG-001 Retest)**	1. Clear cookies 2. Navigate to /admin on local server 3. Verify immediate redirection to /admin/login.	System redirects unauthenticated visitors to /admin/login with security spinner.	Redire http:// strictly
TC-013	Admin Security	**Local Admin API Endpoint Protection (BUG-002 Retest)**	1. Send unauthenticated fetch to /api/admin/orders 2. Verify HTTP response code.	API returns HTTP 401 Unauthorized with error message.	Protec Unauth
TC-014	Admin Authentication	**Admin Login Negative Test (Incorrect Password)**	1. Navigate to /admin/login 2. Enter invalid password "wrong_password_xyz" 3. Submit form.	Form displays error banner "Invalid password" and rejects login.	Error d
TC-015	Admin Authentication	**Admin Login Positive Test & Session Creation**	1. Enter valid admin password "admin" 2. Submit form 3. Verify redirect to /admin dashboard.	Admin cookie set; user redirected to administrative dashboard.	Succe http://
TC-016	Admin Dashboard	**Admin Dashboard Sales Metrics & KPI Cards**	1. View /admin dashboard 2. Verify total revenue KPI, order counters, and active product count.	Dashboard renders live summary statistics from PostgreSQL database.	KPI car metric
TC-017	Admin Orders	**Admin Orders Management Table & Status Filter Tabs**	1. Navigate to /admin/orders 2. Inspect order listing table, status tabs, and logistics columns.	Orders table displays customer names, phone numbers, payment status, and action buttons.	Orders render
TC-018	Admin Products	**Admin Product Catalog Listing & Stock Flags**	1. Navigate to /admin/products 2. Verify product rows, category filters, and bestseller toggles.	Catalog displays fine jewellery products with prices, SKU, stock counters, and edit controls.	Catalo
TC-019	Admin Slider	**Admin Hero Slider Banner Configuration**	1. Navigate to /admin/slider 2. Verify active slides, banner preview images, and sort controls.	Slider manager renders luxury hero banners with active/inactive toggles.	Slider availab

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TC-020	Admin Settings	**Admin Store Settings & Third-Party Integration Configuration**	1. Navigate to /admin/settings 2. Verify Razorpay credentials, Shiprocket settings, and WhatsApp notification parameters.	Settings page displays inputs for API keys, sandboxes, and connection test buttons.	Setting and dia
TC-021	Integrations Diagnostic	**Shiprocket Authentication & API Connectivity Diagnostic**	1. On /admin/settings, click "Test Connection" for Shiprocket 2. Verify API response banner.	System tests API credentials and displays status banner (success or actionable configuration alert).	Shipro creder feedba
TC-022	Integrations Diagnostic	**WhatsApp Business API Alert Dispatch Diagnostic**	1. On /admin/settings, click "Send Test WhatsApp Message" 2. Verify notification dispatch result.	System executes dispatch pipeline and returns success or actionable error message.	Whats throug
TC-023	Admin Catalog Sync	**Admin Google Sheets & Drive Cloud Sync Configuration**	1. Navigate to /admin/sync 2. Verify Google Sheets URL input, sync frequency selector, and manual sync trigger.	Sync configuration renders with sheet URL input and trigger button.	Cloud contro
TC-024	Admin CSV Import	**Admin CSV Catalog Bulk Import & Validation Interface**	1. Navigate to /admin/import 2. Verify file dropzone, CSV template instructions, and validate button.	CSV import interface displays file picker and batch validation workflow.	Import link an
TC-025	Customer Auth	**Customer Registration Tab & Input Fields Rendering**	1. Navigate to /login 2. Click "Create an Account" tab 3. Verify name, phone, email, password, address, and city fields.	Registration form renders with all required customer onboarding inputs.	Regist phone
TC-026	Customer Auth	**Customer Account Creation with Unique Phone & Email**	1. Submit registration with Name: Aarav Sharma, Phone: 9835316123 2. Verify HTTP 200 response and customer token creation.	System creates User record in PostgreSQL and establishes customer session.	User cr Aarav S

TEST ID	MODULE	SCENARIO	EXECUTION STEPS	EXPECTED RESULT	ACTU
TC-027	Customer Auth	**Customer Registration Duplicate Phone Boundary Check**	1. Attempt registration using an already registered mobile number (9876543210) 2. Verify 400 error.	System returns HTTP 400 with "An account with this mobile number already exists".	Reject 400 (A already
TC-028	Customer Auth	**Customer Login Interface & Credential Inputs**	1. Navigate to /login 2. Verify mobile/email identifier input, password input, and Sign In button.	Login interface renders with brand concierge styling and security badges.	Custor creder
TC-029	Customer Auth	**Customer Login Negative Test (Incorrect Password)**	1. Send login POST with invalid password 2. Verify HTTP 401 response.	System returns HTTP 401 with "Incorrect password".	Reject passw
TC-030	Customer Account	**Customer Account Portal Unauthenticated Access Guard**	1. Clear customer cookies 2. Navigate directly to /account 3. Verify redirect to /login?redirect=/account.	Unauthenticated visitor is intercepted and redirected to login page.	Redire redirec
TC-031	Product Details	**Metal Finish & Gold Purity Variant Selection**	1. On PDP, click metal finish option (Yellow Gold / Rose Gold) 2. Verify selected metal visual indicator updates.	Selected metal finish button shows active luxury gold highlight.	Metal v active
TC-032	Cart Management	**Add Product to Cart & Cart Drawer Trigger**	1. Click "Add to Bag" on PDP 2. Verify item appears in cart drawer with title, price, and metal attribute.	Cart drawer opens displaying the added item and updated subtotal.	Item ac price a
TC-033	Checkout Flow	**Empty Cart Checkout Redirection Guard**	1. Clear cart in localStorage 2. Navigate directly to /checkout 3. Verify redirect to catalog/homepage.	User cannot checkout with 0 items; system redirects to product listing.	Empty http://
TC-034	Checkout Flow	**Full Cash on Delivery (COD) Checkout & Navigation (BUG-003 Retest)**	1. Add item to cart 2. Navigate to /checkout 3. Fill valid customer details 4. Click Place Order 5. Verify window.location redirection to /order-success/[orderNumber].	Order created in PostgreSQL, cart cleared atomically, and user routed to order confirmation screen.	Succe http:// false).

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TC-035	Order Confirmation	**Order Success Confirmation & WhatsApp Alert Notice**	1. On order confirmation page, inspect order summary 2. Verify order number, customer name, delivery address, and WhatsApp confirmation notice.	Order confirmation page renders order number, confetti celebration, and WhatsApp notification banner.	Confir (detec (detec
TC-036	Order Tracking	**Track Order with Valid Order Number**	1. On /track-order, enter order number "JM-2026-89640" 2. Click Track Order button 3. Verify order status timeline.	System returns order details, fulfillment stage, items list, and delivery estimation.	Order with ite
TC-037	Order Tracking	**Track Order with Non-Existent Order Number (Negative Test)**	1. Enter fake order ID "INVALID_ORDER_XYZ_999" 2. Click Track Order 3. Verify error message.	System gracefully renders "Order not found" message without crashing.	Error st detect
TC-038	Static Care Pages	**About Us Brand Story & Artisanal Heritage Page**	1. Navigate to /about 2. Verify brand story, craftsman heritage, and gold hallmarking commitments.	About Us page renders with fine typography, images, and brand commitments.	Brand s design
TC-039	Static Care Pages	**Contact Us & Boutique Concierge Page**	1. Navigate to /contact 2. Verify boutique address in Zaveri Bazaar, toll-free number, and contact form.	Contact page renders phone numbers, boutique address, email, and inquiry form.	Concie compl
TC-040	Static Care Pages	**Insured Shipping & Delivery Policy Page**	1. Navigate to /shipping-policy 2. Verify pan-India insured shipping terms and transit delivery timelines.	Shipping policy displays courier insurance, delivery windows, and tamper-proof packaging guarantee.	Shippi insuran
TC-041	Static Care Pages	**15-Day Returns & Lifetime Exchange Policy Page**	1. Navigate to /returns-policy 2. Verify 15-day return window, BIS hallmarking exchange rules, and buyback policy.	Returns policy displays exchange calculation, return eligibility, and free return pickup.	Return terms a

TEST ID	MODULE	SCENARIO	EXECUTION STEPS	EXPECTED RESULT	ACTU
TC-042	Admin Products	**Admin Bestseller Feature Flag Toggle**	1. On /admin/products, click star icon to toggle bestseller flag 2. Verify optimistic UI toggle and PATCH /api/admin/products response.	Product bestseller state updates instantly and persists in PostgreSQL.	Bestse databa
TC-043	Admin Authentication	**Admin Session Logout & Security Cookie Revocation**	1. On /admin, click "Sign Out" button 2. Verify admin cookie deletion and redirection to /admin/login.	System invalidates jm_admin_auth cookie and navigates back to login screen.	Admin http:// (Logge

4. Payment & External Integration Verification

Razorpay Payment Integration

Tested using sandbox test keys (`rzp_test_jewellerymate`). Direct Cash on Delivery (COD) order placement verified transactionally in PostgreSQL without incurring real financial charges.

SANDBOX VERIFIED

Shiprocket Logistics & WhatsApp Alerts

Shiprocket authentication tested via admin diagnostic panel. Meta WhatsApp / Mock notification pipeline verified without sending unsolicited notifications to actual customer mobile numbers.

DIAGNOSTICS VERIFIED